

Chapter 9, Problem 7PP

Problem

Find the voltage $v(t)$ in a circuit described by the integro-differential equation

$$2\frac{dv}{dt} + 5v + 10\int v dt = 50\cos(5t - 30^\circ)$$

using the phasor approach.

Step-by-step solution

Step 1 of 3

Given that

$$2\frac{dv}{dt} + 5v + 10\int v dt = 50\cos(5t - 30^\circ)$$

$$\therefore \omega = 5 \text{ rad/sec}$$

We transform each term in the equation from time domain to phasor domain, we obtain the phasor form of the given equation as

$$2(j\omega V) + 5V + 10\left(\frac{V}{j\omega}\right) = 50\angle -30^\circ \quad \left(\because \frac{dv}{dt} = j\omega V \text{ \& } \int v dt = \frac{V}{j\omega}\right)$$

Step 2 of 3

Substitute $\omega = 5 \text{ rad/sec}$ in the above equation

$$2(j \times 5 \times V) + 5V + 10\left(\frac{V}{j \times 5}\right) = 50\angle -30^\circ$$

$$j10V + 5V + \frac{2V}{j} = 50\angle -30^\circ$$

$$j10V + 5V - j2V = 50\angle -30^\circ \quad \left(\because \frac{1}{j} = -j\right)$$

$$V(j10 + 5 - j2) = 50\angle -30^\circ$$

$$V(5 + j8) = 50\angle -30^\circ \quad \text{.....(1)}$$

Step 3 of 3

By using rectangular to polar transformation

$$5 + j8 = \sqrt{(5)^2 + (8)^2} \angle \tan^{-1}\left(\frac{8}{5}\right)$$

$$5 + j8 = 9.434 \angle 58^\circ$$

Equation (1) can be write as ,

$$V(9.434 \angle 58^\circ) = 50 \angle -30^\circ$$

$$V = \frac{50 \angle -30^\circ}{9.434 \angle 58^\circ}$$

$$V = 5.3 \angle -88^\circ$$

Transforming this to the time domain , we get

$$\boxed{v(t) = 5.3 \cos(5t - 88^\circ) \text{ V}}$$